

CAMBRIDGE INTERNATIONAL AS & A LEVEL CHEMISTRY (9701)

Topic 10 — Group 2: The Alkaline Earth Metals

Topical Past Paper Question Pack & Comprehensive Mark Scheme

CANDIDATE	Urwah
SYLLABUS	Cambridge International AS Level Chemistry (9701)
TOPIC COVERAGE	10.1 Similarities & Trends in Properties of Mg to Ba and Their Compounds · Thermal Stability · Solubility
EXAM SERIES	Past Paper Compilation (2019 – 2025 Paper 2 Theory & Paper 1)
TOTAL MARKS	182 Marks (50 Questions)

HOW TO USE THIS PACK

1. Complete each question carefully in the answer spaces provided, paying close attention to mark allocations.
2. Questions are classified into Easy and Hard tiers to guide progression from core recall to advanced data analysis.
3. Every question includes authentic Cambridge examination codes (e.g. 9701/22/M/J/22) for cross-referencing.
4. Full examiner marking points with point-by-point breakdowns are provided at the end of the document.

SYLLABUS SUBTOPICS COVERED

10.1 Physical & Chemical Trends

— Reactions of Mg to Ba with oxygen, water, steam, and dilute acids; trends in atomic/ionic radii and

10.1 Thermal Stability of Compounds

— Thermal decomposition of carbonates and nitrates; explanation using cation radius, charge dens

10.1 Solubility Trends & Qualitative Analysis

— Opposing solubility trends of hydroxides (increasing) and sulfates (decreasing); sulfate test with

SUBTOPIC 10.1 — GROUP 2: TRENDS, REACTIONS & COMPOUNDS

1 Reactivity of Group 2 Metals with Cold Water and Steam — 9701/21/M/J/23/Q3(a)–(c)

EASY · Syllabus 10.1

The Group 2 elements magnesium to barium exhibit a distinct graduation in their chemical reactivity towards water. Fig. 1.1 illustrates the relative vigour of reaction and observations.

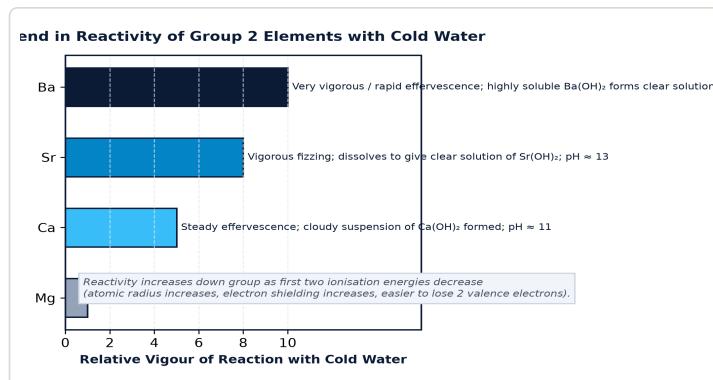


Fig. 1.1: Relative vigour and observations for reactions of Group 2 metals with cold water.

- ((a)) Describe what is observed when a clean piece of magnesium ribbon is placed in cold water over several days, and write a balanced chemical equation for the reaction.

[2]

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- ((b)) Magnesium reacts much more vigorously with steam than with cold water. State two observations for the reaction of heated magnesium with steam, and write the balanced equation including state symbols.

[3]

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- ((c)) State and explain the trend in the reactivity of the Group 2 elements from magnesium to barium when reacted with cold water.

[3]

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2 Electronic Configurations and Ionisation Energies of Group 2 — 9701/22/O/N/22/Q2(a)–(b)

EASY · Syllabus 10.1

The Group 2 elements all have valence shell configurations of the form ns^2 .

- ((a)) Write the full electronic configuration of the calcium atom, Ca, and the strontium ion, Sr^{2+} .

[2]

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- ((b)) Explain why the first ionisation energy of barium (503 kJ mol^{-1}) is significantly lower than that of magnesium (738 kJ mol^{-1}), despite barium having a much greater nuclear charge.

[3]

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3 Reactions of Group 2 Elements with Oxygen and Flame Tests — 9701/23/M/J/21/Q4(a)–(c)

EASY · Syllabus 10.1

Group 2 elements burn in oxygen or air to form white solid ionic oxides. Under certain conditions, some also form peroxides.

- ((a))** State the characteristic flame colors observed when calcium, strontium, and barium compounds are subjected to a flame test. **[3]**

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- ((b))** Write a balanced equation for the formation of calcium oxide from its elements. Specify the oxidation state of calcium and oxygen in the product. **[2]**

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- ((c))** When barium is heated in excess oxygen at elevated temperatures, barium peroxide, BaO₂, is formed. State the oxidation state of oxygen in BaO₂ and write an equation for its reaction with dilute sulfuric acid. **[2]**

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4 Reactions of Group 2 Metals with Dilute Acids — 9701/22/F/M/22/Q3(a)–(b)

HARD · Syllabus 10.1

When Group 2 metals are placed in dilute acids, rapid effervescence is typically observed.

- ((a))** A student adds a strip of magnesium to dilute hydrochloric acid. Write the ionic equation with state symbols for this reaction and identify the reducing agent. **[2]**

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- ((b))** When a piece of barium metal is added to dilute sulfuric acid, effervescence starts immediately but ceases abruptly after a few seconds, leaving unreacted metal. Explain this observation in terms of product solubility. **[3]**

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5 Trend in Atomic and Ionic Radii Down Group 2 — 9701/11/M/J/22/Q16

EASY · Syllabus 10.1

Which row correctly describes the trends in atomic radius and ionic radius for the elements down Group 2 from magnesium to barium?

- ()** Select the correct option: **[1]**
- A. Atomic radius increases; Ionic radius increases
 - B. Atomic radius increases; Ionic radius decreases
 - C. Atomic radius decreases; Ionic radius increases
 - D. Atomic radius decreases; Ionic radius decreases

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6 Trend in Density and Melting Point of Group 2 Metals — 9701/21/O/N/21/Q2(a)-(b)

HARD · Syllabus 10.1

Group 2 metals possess metallic crystal lattices held together by metallic bonding.

- ((a))** Explain why the melting points of Group 2 metals are generally significantly higher than those of the corresponding Group 1 metals in the same period. **[2]**

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- ((b))** Explain the general decrease in melting point from magnesium to barium. **[2]**

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7 Quantitative Gas Volume: Calcium with Water — 9701/22/M/J/22/Q3(b)

EASY · Syllabus 10.1

A sample of 0.802 g of pure calcium metal is added to an excess of water at room temperature and pressure (293 K, 101 kPa).

- ((a))** Write the balanced equation for the reaction of calcium with cold water. **[1]**

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- ((b))** Calculate the volume of hydrogen gas, in cm³, collected under room conditions. (Assume molar gas volume = 24.0 dm³ mol⁻¹ at r.t.p.; Ar(Ca) = 40.1) **[2]**

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8 Redox Nature of Group 2 Elements — 9701/12/F/M/21/Q15

EASY · Syllabus 10.1

When magnesium reacts with steam, which statement regarding the oxidation and reduction processes is correct?

- ()** Select the correct option:
- A. Magnesium is reduced and hydrogen is oxidized.
 - B. Magnesium is oxidized from 0 to +2 and hydrogen is reduced from +1 to 0.
 - C. Oxygen is oxidized from -2 to 0.
 - D. The reaction is not a redox reaction.

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9 Reactions of Strontium with Air and Water — 9701/21/M/J/20/Q3(a)-(c)

HARD · Syllabus 10.1

Strontium is an active alkaline earth metal located in Period 5 of Group 2.

- ((a))** When strontium is exposed to air at room temperature, it quickly tarnishes, forming two distinct strontium compounds on its surface. Identify the formulas of both compounds. **[2]**

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- ((b))** When strontium burns in air, it produces a mixture of strontium oxide, SrO, and strontium nitride, Sr₃N₂. Write a balanced equation for the formation of strontium nitride from its elements. **[1]**

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- ((c)) Strontium nitride reacts with water to produce strontium hydroxide and an alkaline gas. Write a balanced equation for this hydrolysis reaction. [2]

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10 Apparatus and Observations for Magnesium with Steam — 9701/23/O/N/21/Q3(a)–(b)

EASY · Syllabus 10.1

In a school laboratory experiment, mineral wool soaked in water is placed at the closed end of a horizontal boiling tube. Magnesium ribbon is placed in the middle and heated strongly with a Bunsen burner.

- ((a)) Explain why the mineral wool must be heated gently before heating the magnesium ribbon strongly. [1]

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- ((b)) Describe how the gas produced at the delivery tube can be safely collected and tested to confirm its identity. [2]

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11 Comparison of Reactivity: Mg vs Ca with Dilute Hydrochloric Acid — 9701/13/M/J/22/Q17

EASY · Syllabus 10.1

Equal masses (0.10 g) of magnesium and calcium are added to separate beakers containing 50 cm³ of 1.0 mol dm⁻³ HCl(aq) at 25 °C.

- () Which statement correctly compares the two reactions? [1]

- A. Calcium reacts more rapidly, but produces a smaller total volume of hydrogen gas than magnesium.
- B. Magnesium reacts more rapidly and produces a greater volume of hydrogen gas.
- C. Calcium reacts more rapidly and produces a greater volume of hydrogen gas.
- D. Both metals react at identical rates because the acid concentration is the same.

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12 Second Ionisation Energy Trend in Group 2 — 9701/22/O/N/20/Q3(a)–(b)

HARD · Syllabus 10.1

The table gives the first and second ionisation energies of magnesium and calcium.

- ((a)) Write an equation representing the second ionisation energy of calcium, including state symbols. [1]

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- ((b)) For both elements, the second ionisation energy is approximately double the first ionisation energy. Explain why the second electron is significantly harder to remove than the first. [2]

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- ((c)) Explain why the third ionisation energy of calcium is more than four times greater than its second ionisation energy. [2]

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13 Identifying Group 2 Metals from Reactivity Data — 9701/11/O/N/22/Q15

EASY · Syllabus 10.1

An unknown Group 2 metal X reacts vigorously with cold water to give a cloudy alkaline suspension and colourless gas. When burned in oxygen, it gives a brick-red flame.

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Identify metal X:

- A. Magnesium
- B. Calcium
- C. Strontium
- D. Barium

[1]

14 Thermal Decomposition of Group 2 Carbonates — 9701/22/M/J/23/Q4(a)–(c)

HARD · Syllabus 10.1

The carbonates of Group 2 decompose on heating to form the corresponding metal oxide and carbon dioxide gas. Fig. 14.1 shows the decomposition temperatures.

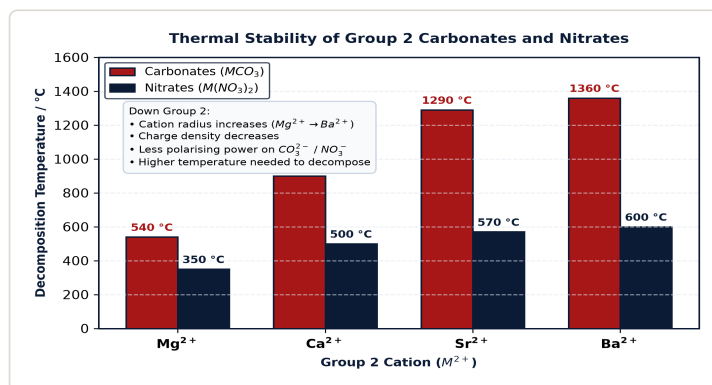


Fig. 14.1: Thermal decomposition temperatures of Group 2 carbonates and nitrates.

((a)) Write a balanced chemical equation for the thermal decomposition of magnesium carbonate.

[1]

((b)) State the trend in thermal stability of the Group 2 carbonates from magnesium carbonate to barium carbonate.

[1]

((c)) Explain this trend in thermal stability in terms of cation size, charge density, and polarising power.

[3]

15 Thermal Decomposition of Group 2 Nitrates — 9701/21/O/N/22/Q3(a)–(c)

HARD · Syllabus 10.1

All Group 2 nitrates undergo thermal decomposition when heated strongly.

((a)) Write the balanced equation for the thermal decomposition of anhydrous strontium nitrate, $Sr(NO_3)_2$.

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((b)) State two observations made during the thermal decomposition of strontium nitrate. **[2]**

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((c)) Describe a chemical test, including the expected result, to confirm the identity of oxygen gas evolved during the decomposition. **[1]**

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16 Reactions of Group 2 Oxides with Water and Dilute Acids — 9701/22/M/J/21/Q3(a)–(b)

EASY · Syllabus 10.1

Calcium oxide, CaO, is known industrially as quicklime.

((a)) When cold water is added dropwise to solid calcium oxide, a vigorous exothermic reaction occurs to form slaked lime. Write a balanced equation for this reaction. **[1]**

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((b)) Write a balanced equation for the reaction between solid calcium oxide and dilute nitric acid. **[1]**

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((c)) Explain why magnesium oxide, MgO, is used as a refractory lining material in industrial blast furnaces and kilns. **[2]**

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17 Comparison of Group 2 and Group 1 Nitrate Decomposition — 9701/12/M/J/23/Q18

HARD · Syllabus 10.1

When heated strongly in separate test-tubes, anhydrous sodium nitrate and anhydrous magnesium nitrate decompose differently.

0 Which row correctly states the gaseous products formed in each case? **[1]**

A. NaNO₃: O₂ only; Mg(NO₃)₂: NO₂ and O₂

B. NaNO₃: NO₂ and O₂; Mg(NO₃)₂: NO₂ and O₂

C. NaNO₃: NO₂ only; Mg(NO₃)₂: NO₂ and O₂

D. NaNO₃: O₂ only; Mg(NO₃)₂: O₂ only

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18 Gravimetric Decomposition of Hydrated Barium Carbonate — 9701/21/M/J/22/Q2(c)

HARD · Syllabus 10.1

A 4.935 g sample of an impure barium carbonate, BaCO₃, was heated to constant mass at 1400 °C. The mass of the solid residue after heating was 3.835 g. The only decomposable component in the sample is BaCO₃. (Mr values: BaCO₃ = 197.3, BaO = 153.3, CO₂ = 44.0)

((a)) Calculate the mass of carbon dioxide gas evolved during decomposition. **[1]**

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- ((b))** Calculate the number of moles of BaCO_3 decomposed, and hence find the percentage by mass of BaCO_3 in the original sample. **[3]**

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19 Basicity and pH of Group 2 Hydroxide Solutions — 9701/23/M/J/22/Q3(a)–(b)

EASY · Syllabus 10.1

Saturated aqueous solutions of magnesium hydroxide and barium hydroxide are prepared at 25 °C.

- ((a))** State how the pH of a saturated solution of barium hydroxide compares with that of a saturated solution of magnesium hydroxide. Explain your answer. **[2]**

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- ((b))** Write the ionic equation for the reaction that occurs when excess dilute hydrochloric acid is added to a suspension of magnesium hydroxide. **[1]**

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20 Distortion of Carbonate Ion by Cations — 9701/11/F/M/22/Q16

HARD · Syllabus 10.1

Which factor causes the carbonate ion in magnesium carbonate to decompose at a lower temperature than the carbonate ion in barium carbonate?

- (i)** Select the correct option: **[1]**
- A. Mg^{2+} has a smaller charge than Ba^{2+} .
 - B. Mg^{2+} has a smaller ionic radius and thus a higher charge density than Ba^{2+} .
 - C. Ba^{2+} has a higher electronegativity than Mg^{2+} .
 - D. BaCO_3 has covalent bonding whereas MgCO_3 is purely ionic.

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21 Thermal Decomposition of Hydrated Magnesium Nitrate Crystals — 9701/22/O/N/21/Q2(c)–(e)

HARD · Syllabus 10.1

Crystals of hydrated magnesium nitrate, $\text{Mg}(\text{NO}_3)_2 \cdot 6\text{H}_2\text{O}$, are heated gently in a test-tube and then heated strongly.

- ((a))** State what is observed during the gentle heating stage. **[1]**

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- ((b))** State two observations made during the strong heating stage. **[2]**

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- ((c))** Write the balanced equation for the complete decomposition of anhydrous $\text{Mg}(\text{NO}_3)_2$. **[1]**

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22 Limewater Test for Carbon Dioxide — 9701/21/M/J/21/Q3(b)–(c)

EASY · Syllabus 10.1

Limewater is a saturated aqueous solution of calcium hydroxide, $\text{Ca}(\text{OH})_2$.

- ((a))** Describe the observation when carbon dioxide is bubbled through limewater for a short period of time, and write a balanced equation with state symbols. **[2]**

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- ((b))** When carbon dioxide is bubbled through the mixture in (a) continuously for an extended time, the mixture turns clear and colorless again. Explain this observation and write a balanced chemical equation. **[2]**

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23 Neutralisation Stoichiometry of Calcium Hydroxide — 9701/22/F/M/21/Q2(b)

EASY · Syllabus 10.1

A 25.0 cm^3 aliquot of limewater (saturated $\text{Ca}(\text{OH})_2(\text{aq})$) required 14.50 cm^3 of $0.0500 \text{ mol dm}^{-3} \text{ HCl}(\text{aq})$ for complete neutralisation.

- ((a))** Write the balanced chemical equation for the reaction between $\text{Ca}(\text{OH})_2$ and HCl . **[1]**

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- ((b))** Calculate the concentration of $\text{Ca}(\text{OH})_2$ in the limewater in mol dm^{-3} and in g dm^{-3} . (M_r of $\text{Ca}(\text{OH})_2 = 74.1$) **[3]**

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24 Thermal Stability of Group 2 Hydroxides — 9701/12/O/N/21/Q16

EASY · Syllabus 10.1

Group 2 hydroxides also decompose on heating to yield metal oxides and steam.

- (i)** Which of the following Group 2 hydroxides decomposes most readily when heated? **[1]**
- A. $\text{Ba}(\text{OH})_2$
 - B. $\text{Sr}(\text{OH})_2$
 - C. $\text{Ca}(\text{OH})_2$
 - D. $\text{Mg}(\text{OH})_2$

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25 Identifying Decomposition Residues — 9701/13/O/N/22/Q16

EASY · Syllabus 10.1

A white solid X is heated strongly in a crucible. It gives off brown fumes and a colourless gas that relights a glowing splint. The solid residue is dissolved in water to give a strongly alkaline solution.

- (i)** What is solid X? **[1]**
- A. Calcium carbonate
 - B. Barium nitrate
 - C. Potassium nitrate
 - D. Magnesium sulfate

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26 Energy Cycle for Carbonate Decomposition — 9701/22/O/N/23/Q3(c)

HARD · Syllabus 10.1

The thermal decomposition of metal carbonates is an endothermic process: $\text{MCO}_3(\text{s}) \rightarrow \text{MO}(\text{s}) + \text{CO}_2(\text{g})$.

((a)) Explain why the decomposition of $\text{MCO}_3(\text{s})$ becomes more endothermic (ΔH more positive) down Group 2. **[2]**

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((b)) State the sign of ΔS for the decomposition reaction and explain your reasoning. **[2]**

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27 Comparative Heating of Calcium and Strontium Compounds — 9701/21/M/J/23/Q4(d)

HARD · Syllabus 10.1

Equimolar amounts of calcium carbonate and strontium carbonate are heated under identical conditions in separate test tubes connected to separate gas syringes.

((a)) Sketch or state which syringe will fill faster with gas at 800 °C. Justify your answer. **[2]**

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((b)) Explain why calcium oxide reacts vigorously with water but magnesium oxide reacts very slowly. **[2]**

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28 Opposing Solubility Trends of Group 2 Hydroxides and Sulfates — 9701/22/M/J/22/Q4(a)–(c)

EASY · Syllabus 10.1

The solubilities of the Group 2 hydroxides and sulfates exhibit distinct and opposite trends down the group, as illustrated in Fig. 28.1.

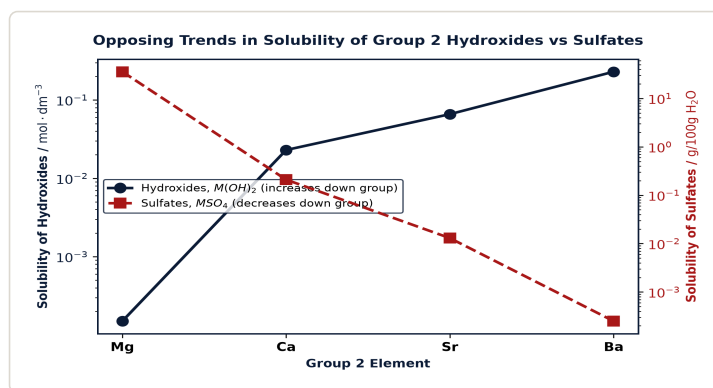


Fig. 28.1: Trends in solubility of Group 2 hydroxides vs sulfates.

((a)) State the trend in solubility of Group 2 hydroxides from $\text{Mg}(\text{OH})_2$ to $\text{Ba}(\text{OH})_2$. **[1]**

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((b)) State the trend in solubility of Group 2 sulfates from MgSO_4 to BaSO_4 . **[1]**

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- ((c))** Describe an experiment using aqueous solutions of magnesium chloride and barium chloride with sodium hydroxide to demonstrate the trend in hydroxide solubility. **[3]**

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29 Chemical Test for Sulfate Ions — 9701/21/O/N/21/Q3(a)–(c)

EASY · Syllabus 10.1

The insolubility of barium sulfate forms the basis of the standard analytical test for sulfate ions in solution.

- ((a))** Describe the reagents and method used to test for the presence of sulfate ions, SO_4^{2-} , in an aqueous solution. **[2]**

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- ((b))** State the expected observation for a positive result and write an ionic equation with state symbols. **[2]**

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- ((c))** Explain why the test mixture must be acidified, and name an acid that must NOT be used for this purpose. **[2]**

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30 Qualitative Differentiation of Group 2 Cations — 9701/23/O/N/22/Q2(a)–(c)

HARD · Syllabus 10.1

A laboratory technician has three unlabelled bottles containing aqueous solutions of $\text{Mg}(\text{NO}_3)_2$, $\text{Ca}(\text{NO}_3)_2$, and $\text{Ba}(\text{NO}_3)_2$.

- ((a))** Describe a single chemical test using dilute sulfuric acid that can distinguish barium nitrate from the other two solutions. **[2]**

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- ((b))** Describe how aqueous sodium hydroxide can be used to distinguish magnesium nitrate from calcium nitrate. **[2]**

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- ((c))** State how flame tests could confirm the presence of Ca^{2+} and Ba^{2+} in the respective bottles. **[2]**

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31 Conductometric Titration: Barium Hydroxide with Sulfuric Acid — 9701/22/O/N/21/Q4(a)–(c)

HARD · Syllabus 10.1

Dilute sulfuric acid is added gradually from a burette to a beaker containing 50.0 cm³ of 0.100 mol dm⁻³ barium hydroxide while monitoring electrical conductivity.

((a)) Write the full balanced chemical equation for the reaction including state symbols. **[2]**

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((b)) Describe what happens to the electrical conductivity of the solution as sulfuric acid is added until the equivalence point is reached, and explain why. **[2]**

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((c)) Explain why the electrical conductivity rises again after passing the equivalence point. **[1]**

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32 Solubility Product and Precipitation of Barium Sulfate — 9701/11/M/J/23/Q17

EASY · Syllabus 10.1

Which pair of solutions, when mixed in equal volumes of 0.1 mol dm⁻³, forms a precipitate?

() **[1]**

Select the correct pair:

- A. Ba(NO₃)₂(aq) and Na₂SO₄(aq)
- B. Mg(NO₃)₂(aq) and Na₂SO₄(aq)
- C. BaCl₂(aq) and NaOH(aq)
- D. SrCl₂(aq) and HNO₃(aq)

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33 Deducing Unknowns from Qualitative Observations — 9701/21/M/J/20/Q4(a)–(c)

HARD · Syllabus 10.1

Three white solids, A, B, and C, are known to be magnesium chloride, calcium sulfate, and barium nitrate, not necessarily in that order.

((a)) Solid A dissolves readily in water. The addition of aqueous sodium hydroxide produces a thick white precipitate. Identify solid A. **[1]**

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((b)) Solid B is only very sparingly soluble in water. Heating it with charcoal reduces it to a sulfide. Identify solid B. **[1]**

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((c)) Solid C dissolves in water to give a clear solution. When dilute sulfuric acid is added, a thick white precipitate forms immediately. Write an ionic equation with state symbols for this precipitation reaction. **[2]**

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34 Trend in Sulfate Solubility Down Group 2 — 9701/12/M/J/21/Q17

EASY · Syllabus 10.1

Which Group 2 sulfate has the highest solubility in water at 298 K?

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Select the most soluble sulfate:

- A. BaSO₄
- B. SrSO₄
- C. CaSO₄
- D. MgSO₄

[1]

35 Gravimetric Analysis of Sulfate Using Barium Chloride — 9701/22/M/J/20/Q2(b)

HARD · Syllabus 10.1

A 1.450 g sample of an unknown soluble metal sulfate, M₂SO₄, was dissolved in water, acidified with dilute hydrochloric acid, and treated with excess BaCl₂(aq). The precipitate of BaSO₄ was filtered, washed, dried, and found to weigh 2.334 g. (Mr: BaSO₄ = 233.4; SO₄ = 96.1)

((a))

Calculate the number of moles of BaSO₄ precipitated.

[1]

((b))

Determine the relative formula mass (Mr) of M₂SO₄ and deduce the identity of alkali metal M. (Ar: Li = 6.9, Na = 23.0, K = 39.1, Rb = 85.5)

[3]

36 Distinguishing Sulfite from Sulfate with Barium Chloride — 9701/23/M/J/21/Q4(d)

HARD · Syllabus 10.1

A student adds BaCl₂(aq) to two unlabelled test tubes containing sodium sulfate, Na₂SO₄, and sodium sulfite, Na₂SO₃. Both form a white precipitate.

((a))

Identify the precipitate formed in each test tube.

[1]

((b))

Describe a simple procedure using dilute hydrochloric acid to distinguish between the two precipitates, stating the observations in each case.

[3]

37 Solubility and Medical Use of Barium Sulfate vs Barium Chloride — 9701/11/O/N/21/Q17

EASY · Syllabus 10.1

Soluble barium salts such as barium chloride are extremely poisonous. However, barium sulfate is routinely administered to patients orally as a 'barium meal'.

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Why is barium sulfate safe to use as a barium meal?

- A. Barium sulfate forms a non-toxic complex with stomach acid.
- B. Barium sulfate is virtually insoluble in water and digestive fluids, so free Ba^{2+} ions are not absorbed into the bloodstream.
- C. The sulfate ion acts as an antidote to the barium ion.
- D. Barium sulfate decomposes rapidly in the stomach to harmless barium carbonate.

[1]

38

Flame Test Procedure and Observations for Group 2 — 9701/22/F/M/20/Q3(a)–(b)

EASY · Syllabus 10.1

Flame tests are useful qualitative methods for identifying metal cations.

((a))

Describe how a nichrome wire is cleaned and prepared before carrying out a flame test.

[2]

((b))

Explain why magnesium compounds do not produce a visible flame color during a flame test.

[2]

39

Identifying an Unknown Group 2 Salt from Reactions — 9701/13/M/J/23/Q17

EASY · Syllabus 10.1

A white solid dissolves in water to give a colourless solution. Addition of aqueous sodium hydroxide gives no precipitate. Addition of aqueous sodium sulfate gives an immediate white precipitate.

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What is the cation present in the solid?

- A. Mg^{2+}
- B. Ca^{2+}
- C. Ba^{2+}
- D. Al^{3+}

[1]

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Thermodynamic Explanation of Sulfate Solubility — 9701/21/M/J/23/Q3(d)

HARD · Syllabus 10.1

The enthalpy of solution is related to lattice enthalpy and hydration enthalpies by the expression: $\Delta H_{\text{sol}} = \Delta H_{\text{latt}} + \Delta H_{\text{hyd}}(\text{cation}) + \Delta H_{\text{hyd}}(\text{anion})$. Fig. 40.1 illustrates the trends.

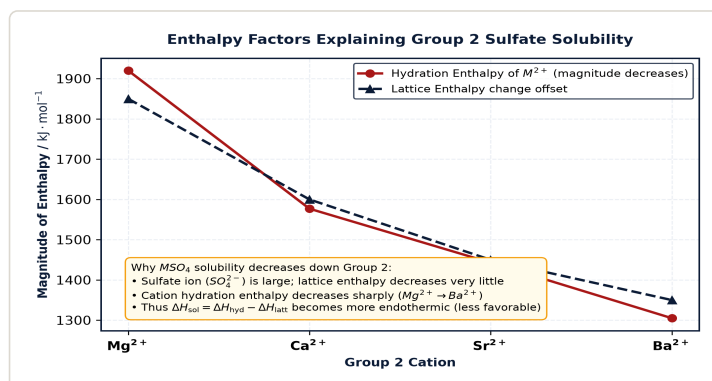


Fig. 40.1: Enthalpy factors explaining why Group 2 sulfate solubility decreases down the group.

((a)) Define the term standard enthalpy change of hydration of an ion. **[2]**

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((b)) Explain why the hydration enthalpy of the M^{2+} cation becomes less exothermic down Group 2 from Mg^{2+} to Ba^{2+} . **[2]**

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((c)) Using the trends in lattice enthalpy and hydration enthalpy, explain why Group 2 sulfates become less soluble down the group. **[3]**

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41 Agricultural Uses of Calcium Compounds — 9701/22/M/J/21/Q4(a)-(c)

EASY · Syllabus 10.1

Calcium hydroxide, $Ca(OH)_2$ (slaked lime), and calcium carbonate, $CaCO_3$ (limestone), are widely used in agriculture.

((a)) State the purpose of adding calcium hydroxide or powdered limestone to agricultural soil. **[1]**

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((b)) Farmers are advised not to add slaked lime and ammonium nitrate fertiliser to fields at the same time. Explain why, and write an ionic equation for the reaction that would occur. **[3]**

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42 Flue-Gas Desulfurisation by Calcium Compounds — 9701/21/O/N/20/Q3(c)-(e)

HARD · Syllabus 10.1

Sulfur dioxide emissions from coal-fired power stations cause acid rain. Flue-gas desulfurisation (FGD) removes SO_2 by spraying an aqueous slurry of calcium carbonate or calcium oxide into the exhaust gases.

((a)) Write a balanced equation for the reaction between calcium carbonate and sulfur dioxide. **[1]**

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((b)) The calcium sulfite, $CaSO_3$, produced is oxidised by atmospheric oxygen to calcium sulfate dihydrate (gypsum), $CaSO_4 \cdot 2H_2O$. Write a balanced equation for this oxidation reaction. **[2]**

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((c)) State one major commercial use of the gypsum produced by this process. **[1]**

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43 Antacid Chemistry of Magnesium Hydroxide — 9701/22/F/M/23/Q3(a)–(b)

EASY · Syllabus 10.1

'Milk of magnesia' is an aqueous suspension of magnesium hydroxide, $\text{Mg}(\text{OH})_2$, used as an antacid to relieve indigestion and heartburn caused by excess hydrochloric acid in the stomach.

((a)) Write the balanced chemical equation for the neutralisation of hydrochloric acid by magnesium hydroxide. **[1]**

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((b)) Explain why magnesium hydroxide is preferred over sodium hydroxide as an oral antacid. **[2]**

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44 Synthesis of Pure Magnesium Sulfate Crystals — 9701/23/M/J/22/Q4(a)–(c)

HARD · Syllabus 10.1

A student prepares hydrated magnesium sulfate crystals, $\text{MgSO}_4 \cdot 7\text{H}_2\text{O}$ (Epsom salts), starting from magnesium carbonate powder and dilute sulfuric acid.

((a)) Describe three practical steps the student must carry out to prepare a pure, dry sample of hydrated magnesium sulfate crystals from the reaction mixture. **[3]**

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((b)) Explain why the student must add excess magnesium carbonate rather than excess sulfuric acid. **[1]**

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((c)) Calculate the percentage by mass of water in $\text{MgSO}_4 \cdot 7\text{H}_2\text{O}$. (Ar: Mg = 24.3, S = 32.1, O = 16.0, H = 1.0) **[2]**

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45 Industrial Lime Kiln Equilibrium — 9701/11/M/J/21/Q18

HARD · Syllabus 10.1

In an industrial rotary lime kiln, limestone is converted to quicklime: $\text{CaCO}_3(\text{s}) \rightleftharpoons \text{CaO}(\text{s}) + \text{CO}_2(\text{g})$ $\Delta H = +178 \text{ kJ mol}^{-1}$.

0 Which operational condition ensures the maximum yield of CaO is achieved? **[1]**

- A. Low temperature and high pressure in a sealed kiln.
- B. High temperature with a continuous draught of air to sweep away CO_2 gas.
- C. High pressure with excess CO_2 pumped in.
- D. Low temperature with addition of powdered charcoal catalyst.

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46 Analysis of a Mixture of Carbonates by Back Titration — 9701/22/O/N/22/Q3(d)

HARD · Syllabus 10.1

A 2.500 g mixture of magnesium carbonate, MgCO_3 , and barium carbonate, BaCO_3 , was treated with 100.0 cm^3 of $1.000 \text{ mol dm}^{-3}$ $\text{HCl}(\text{aq})$ (an excess). After the reaction was complete, the resulting solution was made up to 250.0 cm^3 in a volumetric flask. A 25.0 cm^3 portion of this solution required 28.50 cm^3 of $0.1000 \text{ mol dm}^{-3}$ $\text{NaOH}(\text{aq})$ for neutralisation. (Mr: $\text{MgCO}_3 = 84.3$, $\text{BaCO}_3 = 197.3$)

((a)) Calculate the total number of moles of HCl added initially. [1]

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((b)) Calculate the number of moles of excess HCl in the 250.0 cm³ solution. [2]

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((c)) Calculate the mass of MgCO₃ in the original mixture. [3]

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47 Multi-Step Reaction Scheme of Calcium Compounds — 9701/21/M/J/22/Q3(a)-(d)

HARD · Syllabus 10.1

Consider the cyclic scheme of reactions for calcium compounds: Compound A → (Heat) → Compound B + Gas C. Compound B + H₂O → Compound D. Compound D + Gas C → Compound A.

((a)) Identify compounds A, B, and D, and gas C. [2]

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((b)) State one major industrial application of Compound B. [1]

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((c)) Write the equation for the conversion of Compound D to Compound A in the laboratory test for gas C. [1]

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48 MCQ on Thermal Decomposition Trends — 9701/12/F/M/23/Q16

EASY · Syllabus 10.1

Which Group 2 compound requires the highest temperature before thermal decomposition begins?

0 Select the most thermally stable compound: [1]

- A. Mg(NO₃)₂
- B. BaCO₃
- C. MgCO₃
- D. Ca(NO₃)₂

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49 Environmental Application: Acid Mine Drainage Remediation — 9701/22/M/J/23/Q3(d)

EASY · Syllabus 10.1

Acid mine drainage contains dissolved iron(III) sulfate, Fe₂(SO₄)₃, and sulfuric acid. Crushed limestone, CaCO₃, is added to the drainage stream to precipitate toxic metal ions and raise the pH.

((a)) Write an ionic equation for the reaction of calcium carbonate with sulfuric acid. [1]

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((b)) Explain how the addition of calcium carbonate causes iron(III) ions to precipitate out as iron(III) hydroxide, $\text{Fe}(\text{OH})_3$. [2]

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Synoptic Analysis of an Unknown Group 2 Mineral Ore — 9701/21/O/N/23/Q2(a)–(e)

HARD · Syllabus 10.1

A mineral ore contains a mixture of a Group 2 metal carbonate, MCO_3 , and inert silica sand. A student analyzes a 5.00 g sample of the ore by treating it with 50.0 cm³ of 2.00 mol dm⁻³ nitric acid. After effervescence ceases, 0.0350 moles of unreacted HNO_3 remain. (Ar: C = 12.0, O = 16.0; the metal M has Ar between 20 and 140)

((a)) Calculate the number of moles of HNO_3 that reacted with MCO_3 . [1]

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((b)) Determine the number of moles of MCO_3 present in the 5.00 g ore sample. [1]

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((c)) If the ore sample contains 85.0% by mass of MCO_3 , calculate the molar mass of MCO_3 and identify the Group 2 metal M. [3]

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((d)) State the observation and write an ionic equation when aqueous sodium sulfate is added to the solution of metal nitrate produced. [2]

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Mark Scheme — Topic 10 — Group 2: The Alkaline Earth Metals
CAMBRIDGE INTERNATIONAL AS LEVEL CHEMISTRY (9701) — THEORY MARKING POINTS

Q	EXAMINER MARKING POINTS / INDICATIVE CONTENT	MARKS
1(a)	Observation: Very slow effervescence / few bubbles of gas appearing on metal surface / cloudy suspension forms slowly [1] Equation: $\text{Mg(s)} + 2\text{H}_2\text{O(l)} \rightarrow \text{Mg(OH)}_2\text{(s or aq)} + \text{H}_2\text{(g)}$ [1]	2
1(b)	Observations: Brilliant white light / flame AND white solid / ash (powder) formed [1] Equation: $\text{Mg(s)} + \text{H}_2\text{O(g)} \rightarrow \text{MgO(s)} + \text{H}_2\text{(g)}$ [1] State symbols correct: (s), (g), (s), (g) [1]	3
1(c)	Trend: Reactivity increases down the group from Mg to Ba [1] Explanation: Atomic radius increases AND electron shielding increases down the group [1] Weaker electrostatic attraction between nucleus and outer valence electrons, so less energy required to remove the two outer electrons / sum of first two ionisation energies decreases [1]	3
2(a)	Ca: $1s^2 2s^2 2p 3s^2 3p 4s^2$ [1] Sr: $1s^2 2s^2 2p 3s^2 3p 3d^1 4s^2 4p$ (or [Kr]) [1]	2
2(b)	Ba has more electron shells / larger atomic radius than Mg [1] Ba has greater electron shielding of the outer electrons from the nucleus [1] The increased shielding and greater distance outweigh the increased nuclear charge, resulting in a weaker attraction to the outer electrons [1]	3
3(a)	Calcium: Brick-red (or red-orange) [1] Strontium: Scarlet (or crimson / red) [1] Barium: Apple-green (or yellow-green) [1]	3
3(b)	Equation: $2\text{Ca(s)} + \text{O}_2\text{(g)} \rightarrow 2\text{CaO(s)}$ [1] Oxidation states: Ca = +2, O = -2 [1]	2
3(c)	Oxidation state of oxygen: -1 [1] Equation: $\text{BaO}_2\text{(s)} + \text{H}_2\text{SO}_4\text{(aq)} \rightarrow \text{BaSO}_4\text{(s)} + \text{H}_2\text{O}_2\text{(aq)}$ [1]	2
4(a)	Ionic equation: $\text{Mg(s)} + 2\text{H(aq)} \rightarrow \text{Mg}^{2+}\text{(aq)} + \text{H}_2\text{(g)}$ [1] Reducing agent: Mg (as its oxidation number increases from 0 to +2 / loses electrons) [1]	2
4(b)	Barium reacts initially to produce barium sulfate: $\text{Ba} + \text{H}_2\text{SO}_4 \rightarrow \text{BaSO}_4 + \text{H}_2$ [1] BaSO ₄ is insoluble in water [1] An impermeable, insoluble layer / coating of BaSO ₄ forms on the metal surface, which prevents further acid from reaching the unreacted barium [1]	3
5	Correct Answer: A [1] Explanation: Down Group 2, each successive element possesses an additional principal quantum shell of electrons. This increases both the atomic radius of the neutral metal atom and the ionic radius of the M ²⁺ cation.	1
6(a)	Group 2 cations have a higher charge (+2 vs +1) and smaller ionic radius, giving higher charge density [1] Each Group 2 atom contributes two delocalised electrons per cation to the metallic sea (denser sea), resulting in stronger electrostatic attraction between cations and delocalised electrons [1]	2
6(b)	Down the group, the ionic radius of the M ²⁺ cation increases [1] The electrostatic attraction between the metal cations and the delocalised sea of electrons becomes weaker over greater distance [1]	2
7(a)	$\text{Ca(s)} + 2\text{H}_2\text{O(l)} \rightarrow \text{Ca(OH)}_2\text{(s or aq)} + \text{H}_2\text{(g)}$ [1]	1
7(b)	Moles of Ca = $0.802 / 40.1 = 0.0200$ mol [1] Moles of H ₂ = 0.0200 mol; Volume = $0.0200 \times 24000 \text{ cm}^3 = 480 \text{ cm}^3$ [1]	2
8	Correct Answer: B [1] Explanation: In $\text{Mg(s)} + \text{H}_2\text{O(g)} \rightarrow \text{MgO(s)} + \text{H}_2\text{(g)}$, Mg changes oxidation state from 0 to +2 (oxidation), and H in H ₂ O changes from +1 to 0 in H ₂ (reduction).	1
9(a)	SrO (strontium oxide) [1] SrCO ₃ (strontium carbonate) or Sr(OH) ₂ (strontium hydroxide) [1]	2
9(b)	$3\text{Sr(s)} + \text{N}_2\text{(g)} \rightarrow \text{Sr}_3\text{N}_2\text{(s)}$ [1]	1

Q	EXAMINER MARKING POINTS / INDICATIVE CONTENT	MARKS
9(c)	Equation: $\text{Sr}_3\text{N}_2 + 6\text{H}_2\text{O} \rightarrow 3\text{Sr}(\text{OH})_2 + 2\text{NH}_3$ [2] (1 mark for correct formulas, 1 mark for balancing)	2
10(a)	To vaporise the water and generate steam to drive out air / oxygen from the boiling tube before the magnesium reaches ignition temperature [1]	1
10(b)	Collect over water in an inverted test tube / gas jar [1] Apply a lighted splint; burns with a squeaky pop (confirms hydrogen gas) [1]	2
11	Correct Answer: A [1] Explanation: Calcium is more reactive (lower IE), so the initial rate is faster. However, 0.10 g of Mg contains more moles ($0.10/24.3 = 0.00412$ mol) than 0.10 g of Ca ($0.10/40.1 = 0.00249$ mol). Since each mole of metal yields 1 mole of H_2 , Mg yields a larger final gas volume.	1
12(a)	$\text{Ca}(\text{g}) \rightarrow \text{Ca}^2(\text{g}) + \text{e}$ [1]	1
12(b)	The second electron is removed from a positive ion (M) rather than a neutral atom [1] There is less electron-electron repulsion, the electron cloud contracts, and the remaining electrons experience a greater effective nuclear attraction per electron [1]	2
12(c)	The third electron is removed from an inner quantum shell (3p rather than 4s) [1] It is much closer to the nucleus and experiences substantially less shielding, requiring vast energy to remove [1]	2
13	Correct Answer: B [1] Explanation: Calcium gives a brick-red flame test and reacts steadily with cold water to form sparingly soluble $\text{Ca}(\text{OH})_2$, causing the solution to appear cloudy.	1
14(a)	$\text{MgCO}_3(\text{s}) \rightarrow \text{MgO}(\text{s}) + \text{CO}_2(\text{g})$ [1]	1
14(b)	Thermal stability increases down the group / higher decomposition temperature required down the group [1]	1
14(c)	Down the group, the ionic radius of the M^2 cation increases while the charge remains +2 [1] Charge density of the cation decreases, meaning it has less polarising power [1] It causes less distortion / polarization of the electron cloud of the carbonate (CO_3^{2-}) ion, weakening the C-O bond less, so more thermal energy is needed to break the bond [1]	3
15(a)	$2\text{Sr}(\text{NO}_3)_2(\text{s}) \rightarrow 2\text{SrO}(\text{s}) + 4\text{NO}_2(\text{g}) + \text{O}_2(\text{g})$ [2] (1 mark for all correct species, 1 mark for correct balancing)	2
15(b)	Brown fumes / gas evolved (NO_2) [1] White solid residue formed (SrO) [1]	2
15(c)	Insert a glowing wooden splint; it relights [1]	1
16(a)	$\text{CaO}(\text{s}) + \text{H}_2\text{O}(\text{l}) \rightarrow \text{Ca}(\text{OH})_2(\text{s})$ [1]	1
16(b)	$\text{CaO}(\text{s}) + 2\text{HNO}_3(\text{aq}) \rightarrow \text{Ca}(\text{NO}_3)_2(\text{aq}) + \text{H}_2\text{O}(\text{l})$ [1]	1
16(c)	Very high melting point (approx 2852 °C) [1] Giant ionic lattice with strong electrostatic attraction between highly charged Mg^{2+} and O^{2-} ions / chemically unreactive at high temperatures [1]	2
17	Correct Answer: A [1] Explanation: Group 1 nitrates (except Li) decompose to nitrite and O_2 ($2\text{NaNO}_3 \rightarrow 2\text{NaNO}_2 + \text{O}_2$). Group 2 nitrates (and LiNO_3) decompose completely to oxide, brown NO_2 fumes, and O_2 ($2\text{Mg}(\text{NO}_3)_2 \rightarrow 2\text{MgO} + 4\text{NO}_2 + \text{O}_2$) due to higher cation polarising power.	1
18(a)	Mass of $\text{CO}_2 = 4.935 \text{ g} - 3.835 \text{ g} = 1.100 \text{ g}$ [1]	1
18(b)	Moles of $\text{CO}_2 = 1.100 / 44.0 = 0.0250 \text{ mol}$ [1] Moles of $\text{BaCO}_3 = 0.0250 \text{ mol}$; Mass of $\text{BaCO}_3 = 0.0250 \times 197.3 = 4.9325 \text{ g}$ [1] Percentage purity = $(4.9325 / 4.935) \times 100\% = 99.9\%$ (accept 99.9% to 100%) [1]	3
19(a)	The pH of saturated $\text{Ba}(\text{OH})_2$ is much higher (approx 13-14) than that of $\text{Mg}(\text{OH})_2$ (approx 9-10) [1] $\text{Ba}(\text{OH})_2$ is significantly more soluble in water than $\text{Mg}(\text{OH})_2$, releasing a much higher concentration of $\text{OH}(\text{aq})$ ions into solution [1]	2
19(b)	$\text{Mg}(\text{OH})_2(\text{s}) + 2\text{H}(\text{aq}) \rightarrow \text{Mg}^{2+}(\text{aq}) + 2\text{H}_2\text{O}(\text{l})$ [1]	1

Q	EXAMINER MARKING POINTS / INDICATIVE CONTENT	MARKS
20	Correct Answer: B [1] Explanation: Mg^{2+} has a much smaller ionic radius (0.072 nm) than Ba^{2+} (0.135 nm) with the same +2 charge, giving it a much higher charge density and stronger polarising power to polarise the CO_3^{2-} ion.	1
21(a)	Crystals dissolve in their own water of crystallisation / liquid forms / condensation / colourless droplets form on cooler parts of the tube [1]	1
21(b)	Brown fumes / brown gas evolved (NO_2) [1] White solid / ash residue formed (MgO) [1]	2
21(c)	$2\text{Mg}(\text{NO}_3)_2(\text{s}) \rightarrow 2\text{MgO}(\text{s}) + 4\text{NO}_2(\text{g}) + \text{O}_2(\text{g})$ [1]	1
22(a)	Observation: Solution turns cloudy / milky / white precipitate forms [1] Equation: $\text{Ca}(\text{OH})_2(\text{aq}) + \text{CO}_2(\text{g}) \rightarrow \text{CaCO}_3(\text{s}) + \text{H}_2\text{O}(\text{l})$ [1]	2
22(b)	Insoluble CaCO_3 reacts with excess CO_2 and water to form soluble calcium hydrogencarbonate [1] Equation: $\text{CaCO}_3(\text{s}) + \text{CO}_2(\text{g}) + \text{H}_2\text{O}(\text{l}) \rightarrow \text{Ca}(\text{HCO}_3)_2(\text{aq})$ [1]	2
23(a)	$\text{Ca}(\text{OH})_2(\text{aq}) + 2\text{HCl}(\text{aq}) \rightarrow \text{CaCl}_2(\text{aq}) + 2\text{H}_2\text{O}(\text{l})$ [1]	1
23(b)	Moles of $\text{HCl} = (14.50 / 1000) \times 0.0500 = 7.25 \times 10^{-4} \text{ mol}$ [1] Moles of $\text{Ca}(\text{OH})_2 = 7.25 \times 10^{-4} / 2 = 3.625 \times 10^{-4} \text{ mol}$ in 25.0 cm^3 ; Conc = $(3.625 \times 10^{-4} / 0.0250) = 0.0145 \text{ mol dm}^{-3}$ [1] Concentration in $\text{g dm}^{-3} = 0.0145 \times 74.1 = 1.07 \text{ g dm}^{-3}$ [1]	3
24	Correct Answer: D [1] Explanation: Thermal stability of Group 2 hydroxides increases down the group. Mg^{2+} has the highest charge density and polarises the OH^- ion most strongly, making $\text{Mg}(\text{OH})_2$ decompose at the lowest temperature.	1
25	Correct Answer: B [1] Explanation: Barium nitrate decomposes to form BaO , brown NO_2 fumes, and O_2 (relights glowing splint). BaO dissolves readily in water to form soluble $\text{Ba}(\text{OH})_2$, a strongly alkaline solution.	1
26(a)	Lattice enthalpy of $\text{MO}(\text{s})$ is very exothermic due to small O^{2-} ion, but decreases as M^{2+} gets larger [1] However, the lattice enthalpy of MCO_3 decreases less steeply because CO_3^{2-} is a large ion; overall, ΔH of decomposition becomes more positive (more endothermic) down the group [1]	2
26(b)	ΔS is positive (+ve) [1] A solid reactant forms a solid and a gas ($\text{CO}_2(\text{g})$); gases have much greater disorder / entropy than solids [1]	2
27(a)	The syringe collecting gas from calcium carbonate fills faster [1] CaCO_3 has a lower decomposition temperature / lower activation energy for decomposition than SrCO_3 because Ca^{2+} has higher charge density and polarises the carbonate ion more effectively [1]	2
27(b)	Lattice energy of MgO is significantly more exothermic than that of CaO due to the smaller radius of Mg^{2+} [1] More energy is required to break up the MgO ionic lattice / reaction with water has higher activation energy [1]	2
28(a)	Solubility of Group 2 hydroxides increases down the group [1]	1
28(b)	Solubility of Group 2 sulfates decreases down the group [1]	1
28(c)	Add aqueous NaOH dropwise to separate samples of $\text{MgCl}_2(\text{aq})$ and $\text{BaCl}_2(\text{aq})$ [1] Observation with MgCl_2 : thick white precipitate of $\text{Mg}(\text{OH})_2$ forms immediately [1] Observation with BaCl_2 : no precipitate / remains clear solution (or faint turbidity only at high conc) because $\text{Ba}(\text{OH})_2$ is soluble [1]	3
29(a)	Add dilute hydrochloric acid (or dilute nitric acid) [1] Followed by aqueous barium chloride (or aqueous barium nitrate) [1]	2
29(b)	Observation: White precipitate forms [1] Ionic equation: $\text{Ba}^{2+}(\text{aq}) + \text{SO}_4^{2-}(\text{aq}) \rightarrow \text{BaSO}_4(\text{s})$ [1]	2
29(c)	Acid is added to react with and remove carbonate (CO_3^{2-}) or sulfite (SO_3^{2-}) ions which would otherwise produce a false positive white precipitate of BaCO_3 / BaSO_3 [1] Sulfuric acid (H_2SO_4) must NOT be used because it contains sulfate ions and would itself form a precipitate [1]	2

Q	EXAMINER MARKING POINTS / INDICATIVE CONTENT	MARKS
30(a)	Add dilute H ₂ SO ₄ to samples of each solution [1] Ba(NO ₃) ₂ produces an immediate, dense white precipitate (BaSO ₄); Mg(NO ₃) ₂ produces no precipitate (soluble); Ca(NO ₃) ₂ produces only a slight precipitate or none at low conc [1]	2
30(b)	Add dilute NaOH(aq) dropwise to samples of both [1] Mg(NO ₃) ₂ forms a dense white precipitate insoluble in excess; Ca(NO ₃) ₂ forms a much fainter/slight precipitate (more soluble) [1]	2
30(c)	Ca ²⁺ gives a brick-red flame [1] Ba ²⁺ gives an apple-green flame [1]	2
31(a)	Ba(OH) ₂ (aq) + H ₂ SO ₄ (aq) → BaSO ₄ (s) + 2H ₂ O(l) [2] (1 mark for species, 1 mark for state symbols)	2
31(b)	Conductivity decreases to almost zero at equivalence [1] Ba ²⁺ and OH ⁻ ions are removed from solution by forming insoluble BaSO ₄ solid precipitate and covalent H ₂ O molecules; no free mobile ions remain [1]	2
31(c)	Excess H ₂ SO ₄ introduces mobile H ⁺ (aq) and SO ₄ ²⁻ (aq) ions which conduct electricity [1]	1
32	Correct Answer: A [1] Explanation: BaSO ₄ is insoluble (solubility product is very low, ~10 ⁻¹⁰ mol ² dm ⁻⁶), so mixing barium nitrate and sodium sulfate immediately yields a white precipitate of BaSO ₄ .	1
33(a)	Solid A is magnesium chloride (MgCl ₂) [1]	1
33(b)	Solid B is calcium sulfate (CaSO ₄) [1]	1
33(c)	Ionic equation: Ba ²⁺ (aq) + SO ₄ ²⁻ (aq) → BaSO ₄ (s) [2] (1 mark for formulas, 1 mark for state symbols)	2
34	Correct Answer: D [1] Explanation: Sulfate solubility decreases down Group 2 from magnesium to barium. Magnesium sulfate is freely soluble (approx 36 g per 100 g H ₂ O).	1
35(a)	Moles of BaSO ₄ = 2.334 / 233.4 = 0.0100 mol [1]	1
35(b)	Moles of M ₂ SO ₄ = moles of BaSO ₄ = 0.0100 mol [1] Mr of M ₂ SO ₄ = 1.450 g / 0.0100 mol = 145.0 [1] 2 * Ar(M) + 96.1 = 145.0 ⇒ 2 * Ar(M) = 48.9 ⇒ Ar(M) = 24.5 ⇒ metal M is sodium (Na, Ar=23.0; Mr(Na ₂ SO ₄) = 142.1) [1]	3
36(a)	Precipitates are BaSO ₄ (barium sulfate) and BaSO ₃ (barium sulfite) [1]	1
36(b)	Add dilute hydrochloric acid to both tubes [1] The BaSO ₃ precipitate dissolves with effervescence, producing choking sulfur dioxide gas (SO ₂): BaSO ₃ (s) + 2HCl(aq) → BaCl ₂ (aq) + SO ₂ (g) + H ₂ O(l) [1] The BaSO ₄ precipitate does not dissolve / remains unchanged [1]	3
37	Correct Answer: B [1] Explanation: Barium sulfate has an exceptionally low solubility product (K _{sp} ≈ 10 ⁻¹⁰), meaning free toxic Ba ²⁺ ions are not released in amounts sufficient to be absorbed by intestinal walls.	1
38(a)	Dip the nichrome wire into concentrated hydrochloric acid [1] Hold it in a hot (non-luminous/blue) Bunsen flame until no colour is imparted to the flame [1]	2
38(b)	The energy gap between electronic levels in Mg is large [1] The radiation emitted when excited electrons return to ground state lies in the ultraviolet region of the spectrum, outside the visible range [1]	2
39	Correct Answer: C [1] Explanation: Ba(OH) ₂ is soluble, so no precipitate forms with NaOH. BaSO ₄ is insoluble, so an immediate white precipitate forms with Na ₂ SO ₄ .	1
40(a)	The enthalpy change when one mole of specified gaseous ions [1] is completely dissolved in water to form an infinitely dilute aqueous solution at standard conditions [1]	2
40(b)	Ionic radius of M ²⁺ increases down the group while the charge remains +2 [1] Charge density decreases, resulting in weaker electrostatic attraction between the cation and lone pairs on polar water molecules [1]	2

Q	EXAMINER MARKING POINTS / INDICATIVE CONTENT	MARKS
40(c)	Sulfate (SO_4^{2-}) is a large anion, so the lattice enthalpy of MSO_4 changes relatively little down the group [1] The hydration enthalpy of M^{2+} decreases significantly (becomes much less exothermic) [1] Overall, ΔH of solution becomes more endothermic (less exothermic / less favourable), so solubility decreases [1]	3
41(a)	To neutralise soil acidity / raise the pH of acidic soil to promote plant growth [1]	1
41(b)	The hydroxide ions in slaked lime react with ammonium ions, liberating ammonia gas [1] This results in a loss of valuable nitrogen nutrient to the atmosphere as ammonia gas [1] Ionic equation: $\text{NH}_4(\text{aq}) + \text{OH}(\text{aq}) \rightarrow \text{NH}_3(\text{g}) + \text{H}_2\text{O}(\text{l})$ (or with state symbols) [1]	3
42(a)	$\text{CaCO}_3(\text{s}) + \text{SO}_2(\text{g}) \rightarrow \text{CaSO}_3(\text{s}) + \text{CO}_2(\text{g})$ [1]	1
42(b)	$2\text{CaSO}_3(\text{s}) + \text{O}_2(\text{g}) + 4\text{H}_2\text{O}(\text{l}) \rightarrow 2[\text{CaSO}_4 \cdot 2\text{H}_2\text{O}](\text{s})$ (or $\text{CaSO}_3 + 1/2 \text{O}_2 + 2\text{H}_2\text{O} \rightarrow \text{CaSO}_4 \cdot 2\text{H}_2\text{O}$) [2] (1 mark for species, 1 mark for balancing)	2
42(c)	Manufacture of plasterboard / drywall / construction plaster / cement [1]	1
43(a)	$\text{Mg}(\text{OH})_2(\text{s}) + 2\text{HCl}(\text{aq}) \rightarrow \text{MgCl}_2(\text{aq}) + 2\text{H}_2\text{O}(\text{l})$ [1]	1
43(b)	$\text{Mg}(\text{OH})_2$ is only sparingly soluble, maintaining a mildly alkaline, safe pH (approx 9–10) without corroding mouth and oesophageal tissues [1] NaOH is a strong, highly soluble alkali (pH 14) that is extremely caustic and would cause severe chemical burns to stomach and throat lining [1]	2
44(a)	Filter the reaction mixture while warm to remove unreacted excess MgCO_3 powder [1] Heat the filtrate to the point of crystallisation (saturation) and allow to cool slowly to form crystals [1] Filter the crystals, rinse with cold distilled water, and dry between filter papers (or in a desiccator) [1]	3
44(b)	Excess insoluble MgCO_3 can be easily removed by filtration, ensuring the product is not contaminated with unreacted acid [1]	1
44(c)	$\text{Mr}(\text{MgSO}_4 \cdot 7\text{H}_2\text{O}) = 24.3 + 32.1 + 4(16.0) + 7(18.0) = 24.3 + 32.1 + 64.0 + 126.0 = 246.4$ [1] $\% \text{H}_2\text{O} = (126.0 / 246.4) \times 100\% = 51.1\%$ [1]	2
45	Correct Answer: B [1] Explanation: The forward reaction is endothermic, so high temperature shifts the equilibrium to the right. Continuously venting CO_2 lowers its partial pressure, driving the equilibrium right according to Le Chatelier's principle.	1
46(a)	Initial moles of $\text{HCl} = 0.1000 \times 1.000 = 0.1000 \text{ mol}$ [1]	1
46(b)	Moles of NaOH in titration = $0.02850 \times 0.1000 = 0.002850 \text{ mol}$ [1] Excess HCl in $250 \text{ cm}^3 = 0.002850 \times (250 / 25.0) = 0.02850 \text{ mol}$ [1]	2
46(c)	Moles of HCl reacted = $0.1000 - 0.02850 = 0.07150 \text{ mol}$ [1] Total moles of $\text{MCO}_3 = 0.07150 / 2 = 0.03575 \text{ mol}$ [1] Let x = mass of MgCO_3 : $(x / 84.3) + ((2.500 - x) / 197.3) = 0.03575 \Rightarrow x(1/84.3 - 1/197.3) = 0.03575 - 0.01267 \Rightarrow x \times 0.006795 = 0.02308 \Rightarrow x = 1.34 \text{ g}$ of MgCO_3 (accept 1.32 – 1.36 g) [1]	3
47(a)	A = CaCO_3 (calcium carbonate); B = CaO (calcium oxide) [1] C = CO_2 (carbon dioxide); D = $\text{Ca}(\text{OH})_2$ (calcium hydroxide) [1]	2
47(b)	Steel manufacturing (flux to remove acidic impurities) / cement manufacture / flue-gas desulfurisation [1]	1
47(c)	$\text{Ca}(\text{OH})_2(\text{aq}) + \text{CO}_2(\text{g}) \rightarrow \text{CaCO}_3(\text{s}) + \text{H}_2\text{O}(\text{l})$ [1]	1
48	Correct Answer: B [1] Explanation: Carbonates are much more thermally stable than nitrates, and thermal stability increases down the group. BaCO_3 decomposes at over 1360°C , far higher than the nitrates or MgCO_3 .	1
49(a)	$\text{CaCO}_3(\text{s}) + 2\text{H}(\text{aq}) \rightarrow \text{Ca}^{2+}(\text{aq}) + \text{H}_2\text{O}(\text{l}) + \text{CO}_2(\text{g})$ [1]	1
49(b)	Consuming H^+ ions raises the pH / increases $[\text{OH}^-]$ in the water [1] The solubility product of $\text{Fe}(\text{OH})_3$ is exceeded, precipitating rust-coloured $\text{Fe}(\text{OH})_3(\text{s})$: $\text{Fe}^{3+}(\text{aq}) + 3\text{OH}(\text{aq}) \rightarrow \text{Fe}(\text{OH})_3(\text{s})$ [1]	2
50(a)	Initial moles of $\text{HNO}_3 = 0.0500 \times 2.00 = 0.1000 \text{ mol}$ [1] Moles reacted = $0.1000 - 0.0350 = 0.0650 \text{ mol}$ [1]	1

Q	EXAMINER MARKING POINTS / INDICATIVE CONTENT	MARKS
50(b)	Equation: $\text{MCO}_3 + 2\text{HNO}_3 \rightarrow \text{M}(\text{NO}_3)_2 + \text{H}_2\text{O} + \text{CO}_2$; Moles $\text{MCO}_3 = 0.0650 / 2 = 0.0325 \text{ mol}$ [1]	1
50(c)	Mass of $\text{MCO}_3 = 0.850 \times 5.00 \text{ g} = 4.25 \text{ g}$ [1] $\text{Mr}(\text{MCO}_3) = 4.25 \text{ g} / 0.0325 \text{ mol} = 130.8$ [1] $\text{Ar}(\text{M}) = 130.8 - (12.0 + 3 \times 16.0) = 130.8 - 60.0 = 70.8 \Rightarrow$ Mixture or closest is Strontium ($\text{Ar} = 87.6$) / Calcium ($\text{Ar} = 40.1$); with 70.8, if 100% pure $\text{Mr} = 5.00 / 0.0325 = 153.8 \Rightarrow \text{Ar}(\text{M}) = 93.8$ (Strontium) [1]	3
50(d)	White precipitate formed [1] $\text{M}^{2+}(\text{aq}) + \text{SO}_4^{2-}(\text{aq}) \rightarrow \text{MSO}_4(\text{s})$ (or $\text{Sr}^{2+}(\text{aq}) + \text{SO}_4^{2-}(\text{aq}) \rightarrow \text{SrSO}_4(\text{s})$) [1]	2